

SRS Part 3

SRS Part 3 — System Features (5.8-5.13), Other Nonfunctional Requirements, Other Requirements, Appendices

Role-Specific AI Voice Assistant Framework for Organizations and Entities — Jackfruit Phase-1, Team NexGen (Team ID 43)

5. System Features (continued)

5.8 Integration Management

5.8.1 Description and Priority This feature allows an organization administrator to configure and manage connectors to internal organizational systems (such as HRMS, ITSM, or CRM/admissions systems) that role-assistant workflows can invoke. *Priority: Medium — Benefit: 7, Penalty: 5, Cost: 7, Risk: 6*

5.8.2 Stimulus/Response Sequences - Administrator selects “Manage Integrations” → System displays the list of currently configured integrations. - Administrator adds a new integration by providing connection details (endpoint, authentication, supported operations) → System validates and stores the integration configuration. - Administrator tests an integration → System attempts a sample call and displays the result. - Administrator removes or disables an integration → System prevents workflows from invoking that integration going forward.

5.8.3 Functional Requirements

ID	Requirement
REQ-8.1	The system shall allow an authorized administrator to configure a connector to an internal organizational system.
REQ-8.2	The system shall validate integration configuration details before activation.
REQ-8.3	The system shall allow an administrator to test a configured integration and view the result.
REQ-8.4	The system shall allow an administrator to disable or remove an existing integration.
REQ-8.5	The system shall prevent workflows from invoking an integration that has been disabled or removed.

5.9 Escalation to Human Agent

5.9.1 Description and Priority This feature routes a query to a human agent when the assistant cannot recognize the intent with sufficient confidence or cannot successfully complete the mapped workflow, ensuring the user's request is still addressed. *Priority: High* — *Benefit: 8, Penalty: 7, Cost: 5, Risk: 4*

5.9.2 Stimulus/Response Sequences - A query is flagged for escalation → System adds the conversation to the escalation queue and notifies available human agents. - A human agent accepts an escalated conversation → System assigns the conversation to that agent and displays the relevant context. - The human agent responds to the user → System delivers the agent's response to the user through the conversation interface. - No human agent is available → System notifies the user of the delay and keeps the conversation in the queue.

5.9.3 Functional Requirements

ID	Requirement
REQ-9.1	The system shall route a query to the escalation queue when intent confidence is below the configured threshold or the mapped workflow fails.
REQ-9.2	The system shall notify available human agents of a new escalated conversation.
REQ-9.3	The system shall allow a human agent to accept and respond to an escalated conversation.
REQ-9.4	The system shall deliver the human agent's response to the user through the conversation interface.
REQ-9.5	The system shall notify the user when no human agent is immediately available.

5.10 Conversation History and Notifications

5.10.1 Description and Priority This feature allows users to view their past conversations with role assistants and receive notifications regarding escalation and resolution events. *Priority: Medium* — *Benefit: 7, Penalty: 5, Cost: 4, Risk: 2*

5.10.2 Stimulus/Response Sequences - User opens "My Conversations" → System displays a list of the user's past conversations with relevant role assistants. - User selects a specific conversation → System displays the full transcript and outcome. - A conversation is escalated or resolved → System generates a notification for the user. - User opens a notification → System displays the related conversation and its current status.

5.10.3 Functional Requirements

ID	Requirement
REQ-10.1	The system shall store the transcript and outcome of every conversation between a user and a role assistant. The system shall allow a user to

REQ-10.2	view a list of their past conversations.
REQ-10.3	The system shall display the full transcript and outcome when a user selects a specific conversation.
REQ-10.4	The system shall generate a notification for the user when a conversation is escalated or resolved.

5.11 Admin Dashboard and Monitoring

5.11.1 Description and Priority This feature provides administrators with a central interface to monitor role assistant usage, escalation activity, and overall system health. *Priority: Medium — Benefit: 7, Penalty: 5, Cost: 5, Risk: 3*

5.11.2 Stimulus/Response Sequences - Administrator logs in → System authenticates the administrator and displays the admin dashboard. - Administrator opens the dashboard → System displays summary statistics such as active role assistants, conversation volume, and escalation rate. - Administrator selects a role assistant → System displays usage details specific to that role. - An unauthorized user attempts to access the dashboard → System denies access and displays an appropriate error.

5.11.3 Functional Requirements

ID	Requirement
REQ-11.1	The system shall provide an administrator dashboard summarizing role assistant usage and status.
REQ-11.2	The system shall display conversation volume and escalation rate statistics on the dashboard.
REQ-11.3	The system shall allow an administrator to drill down into usage details for a specific role assistant.
REQ-11.4	The system shall restrict access to the administrator dashboard to authenticated and authorized administrator accounts.

5.12 Reports and Analytics

5.12.1 Description and Priority This feature allows administrators to generate and view reports on intents handled, escalation frequency, and role-wise usage, supporting continuous improvement of role configurations. *Priority: Low — Benefit: 6, Penalty: 3, Cost: 6, Risk: 4*

5.12.2 Stimulus/Response Sequences - Administrator opens “Reports and Analytics” → System retrieves available conversation data and displays relevant reports. - Administrator applies a filter (e.g., by role, date range, or intent) → System processes the data and displays the filtered results. - Administrator requests updated analytics → System refreshes the report using the latest available data.

5.12.3 Functional Requirements

ID	Requirement
REQ-12.1	The system shall allow authorized administrators to generate reports based on available conversation and escalation data.
REQ-12.2	The system shall provide analytics on intent frequency, escalation rate, and role-wise usage.
REQ-12.3	The system shall allow administrators to filter reports by role, date range, or intent.
REQ-12.4	The system shall present generated reports and analytics in a clear and understandable format.

5.13 Role-Based Access Control

5.13.1 Description and Priority This feature ensures that users, administrators, and human agents can access only the system functions and information authorized for their respective roles.
Priority: High — Benefit: 9, Penalty: 8, Cost: 4, Risk: 5

5.13.2 Stimulus/Response Sequences - A user, administrator, or human agent logs in → System authenticates the account and identifies its authorized role. - A user accesses the system → System provides access only to the functions permitted for the user role (role selection, conversation, history, escalation). - An administrator accesses the system → System provides access to authorized administrative functions (role configuration, integration management, dashboard, reports). - A user attempts to access an administrator-only or agent-only function → System denies access and displays an appropriate authorization error.

5.13.3 Functional Requirements

ID	Requirement
REQ-13.1	The system shall support role-based access control for users, administrators, and human agents.
REQ-13.2	The system shall restrict users to the functions and data authorized for their role.
REQ-13.3	The system shall provide administrators with access only to authorized administrative functions.
REQ-13.4	The system shall provide human agents with access only to the escalation queue and related conversation data.
REQ-13.5	The system shall enforce authorization checks before allowing access to protected system functions and data.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- **NFR-P1:** The system shall convert captured voice input to text within a reasonable response time under the typical prototype workload.
- **NFR-P2:** The system shall recognize the intent of a query and return a response within a reasonable response time under the typical dataset size used during testing and demonstration.
- **NFR-P3:** The administrator dashboard shall load usage statistics within a reasonable response time under the expected prototype workload.
- **NFR-P4:** Reports and analytics operations shall return results within a reasonable response time under the expected prototype workload.
- **NFR-P5:** Escalated conversations shall be delivered to the notification queue for human agents without unnecessary delay.

6.2 Safety Requirements

The Role-Specific AI Voice Assistant Framework is a web-based information and workflow-automation system and does not directly control physical equipment or safety-critical processes. - **NFR-S1:** The system shall prevent accidental loss or modification of role-profile, workflow, and conversation data by validating important configuration operations before they are processed. - **NFR-S2:** The system shall preserve conversation and escalation information required for tracking, auditing, and analytics. - **NFR-S3:** The system shall display appropriate error messages when invalid or incomplete configuration or query information is entered so that users and administrators can correct it before submission. - **NFR-S4:** The system shall require confirmation for important configuration operations, such as deleting a role profile or disabling an integration, where accidental actions could affect assistant availability. - **NFR-S5:** No formal safety certification is required for the current academic prototype because the system does not operate or control safety-critical physical systems.

6.3 Security Requirements

- **NFR-SEC1:** The system shall require users, administrators, and human agents to authenticate using valid account credentials before accessing protected system functions.
- **NFR-SEC2:** The system shall protect user authentication credentials and shall not store passwords in plain text.
- **NFR-SEC3:** The system shall use secure communication mechanisms such as HTTPS during deployment to protect user credentials, voice data, and conversation content transmitted between the client and server.
- **NFR-SEC4:** The system shall enforce role-based access control at the application level so that users cannot access administrator-only or agent-only functions.
- **NFR-SEC5:** The system shall restrict access to personal information and conversation content to authorized users, administrators, and human agents according to their roles.
- **NFR-SEC6:** The system shall prevent unauthenticated users from accessing protected system functions.
- **NFR-SEC7:** The system shall maintain records of relevant administrative activities, including role profile changes, integration configuration changes, and escalation handling, to support accountability.
- **NFR-SEC8:** The system shall follow appropriate security and

privacy practices for protecting user and conversation information within the scope of the academic prototype.

6.4 Software Quality Attributes

- **Usability:** The system shall provide a simple, clear, and consistent voice/chat interface that allows users to converse with a role assistant without requiring specialized training.
- **Availability:** The system shall be available and functional during scheduled testing, demonstration, and evaluation periods.
- **Reliability:** The system shall handle invalid or unrecognized inputs and unexpected application conditions gracefully without crashing or causing loss of conversation data.
- **Scalability:** The system shall use a design that allows additional role profiles and integrations to be added without redesigning the core conversational engine.
- **Maintainability:** The system shall use a structured codebase with clearly separated speech processing, intent recognition, workflow execution, and data-access components so that individual features can be modified without unnecessarily affecting unrelated functionality.
- **Data Integrity:** The system shall maintain accurate and consistent role-profile, conversation, escalation, and integration data throughout the conversation lifecycle.
- **Accuracy:** The system shall provide sufficiently accurate intent recognition results for the intents configured during system testing.
- **Fault Tolerance:** The system shall handle speech-processing and integration errors gracefully and shall provide appropriate fallback responses without terminating the entire application.
- **Compatibility:** The web application shall function correctly on commonly used modern web browsers that support microphone access, including Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **Auditability:** The system shall maintain relevant records of role-profile changes, integration changes, and escalation handling to support traceability and accountability.
- **Privacy:** The system shall limit access to personal information and conversation content to authorized users, administrators, and human agents and shall avoid unnecessary exposure of such information.
- **Accessibility:** The system shall provide readable text, clear labels, understandable navigation, and voice interaction to support users with different levels of digital familiarity.
- **Relative Priority:** For this system, accurate intent recognition and reliable escalation to a human agent are prioritized over advanced customization, since users must be able to get a correct answer or reach a human when the assistant cannot help.

6.5 Business Rules

- **BR1:** Only registered and authenticated users shall be permitted to converse with a role assistant.
- **BR2:** Users shall be permitted to view and access only their own conversation history.
- **BR3:** Only authorized administrators shall be permitted to create, edit, or deactivate role profiles.
- **BR4:** Only authorized administrators shall be permitted to configure or manage integrations with internal organizational systems.
- **BR5:** A query shall be escalated to a human agent whenever intent confidence falls below the configured threshold or the mapped workflow cannot be completed.
- **BR6:** Escalated conversations shall be made available only to

- authorized human agents for handling.
- **BR7:** Only authorized administrators shall be permitted to access the administrative dashboard and reports and analytics.
- **BR8:** The system shall maintain each conversation as an individual record and shall preserve its relevant information throughout the conversation lifecycle.
- **BR9:** Conversation and user information shall be accessible only according to the user's authorized role.

Domain Requirements - DR1: Each role assistant shall operate using a distinct set of intents and workflow mappings specific to its organizational role. - **DR2:** Each conversation shall maintain information about its current status (in progress, resolved, escalated) so that its progress can be tracked. - **DR3:** A query that cannot be resolved by the assistant shall be capable of being escalated to a human agent. - **DR4:** Workflow execution shall be capable of invoking configured internal-tool integrations to complete a user request. - **DR5:** Personal information exchanged during a conversation shall be protected and shall not be unnecessarily exposed to unauthorized users.

7. Other Requirements

- **Legal/Compliance Requirement:** The system shall clearly state that it is an academic prototype developed as part of the Jackfruit mini-project and is not an official production assistant of any organization.
 - **Scope Exclusion Requirement:** The system shall clearly indicate that requests requiring legal, medical, or emergency response are outside the scope of the assistant and should be directed to the appropriate authority or emergency service.
 - **Data Retention Requirement:** The system shall retain conversation records and related information for as long as required for tracking, escalation handling, reporting, and system evaluation.
 - **Database Requirement:** The system shall use a MySQL database to store user information, role profiles, intents, workflow mappings, integration configuration, conversation history, escalation records, and analytics data.
 - **Data Backup Requirement:** The system should support appropriate backup and recovery procedures for preserving role-profile, integration, and conversation data.
 - **Reuse Requirement:** Common components such as the conversation interface, authentication forms, and notification components should be designed for reuse across all role assistants to improve consistency and reduce development effort.
 - **Future Enhancement Requirement:** Possible future enhancements may include support for additional languages, sentiment-aware responses, proactive notifications, and integration with real production HRMS, ITSM, and CRM systems. These enhancements are outside the scope of the current project.
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Appendix A: Glossary

Term	Definition
Role Assistant	A configured instance of the AI voice assistant tailored to a specific organizational role (e.g., HR, IT Support, Admissions).
	A configuration record defining a

Role Profile	role assistant's name, supported intents, sample utterances, and workflow mappings.
Intent	The purpose or goal behind a user's query, recognized by the NLU engine (e.g., "check leave balance").
Entity	A specific piece of information extracted from a user query that is required to execute a workflow (e.g., a date or ticket category).
Workflow	A predefined sequence of actions, including internal-tool integration calls, executed in response to a recognized intent.
Escalation	The process of routing a query to a human agent when the assistant cannot resolve it with sufficient confidence.
NLU	Natural Language Understanding – the component responsible for intent recognition and entity extraction.
STT	Speech-to-Text – the component that converts spoken audio into text.
TTS	Text-to-Speech – the component that converts text responses into spoken audio.
Integration Connector	A configurable module that allows the workflow automation engine to call an internal organizational system's API.
Role-Based Access Control (RBAC)	A security mechanism that restricts system functions and information according to the user's authorized role.
Human Agent	Support staff who receive and resolve conversations escalated by the assistant.
Flask/FastAPI	The Python-based web application framework used to develop the backend of the system.
MySQL	The relational database management system used to store and manage system data.

Appendix B: Field Layouts

Sample field layout for the Role Profile / Intent registration data, to be maintained as a companion spreadsheet containing field layouts, properties/attributes, and report requirements.

Field	Length	Data Type	Description	M
Role Name	50	String	Name of the role assistant (e.g., HR	Y

			Assistant)	
Intent Name	50	String	Unique name of the intent within the role	Y
Sample Utterances	255	Text	Comma-separated example phrases for the intent	Y
Workflow ID	16	Alphanumeric	Identifier of the workflow mapped to the intent	Y
Integration Endpoint	255	String	URL/endpoint of the internal system integration, if applicable	N
Confidence Threshold	5	Numeric	Minimum confidence score required to avoid escalation	Y
Conversation ID	16	Alphanumeric	Unique identifier of a conversation record	Y
Escalation Status	20	Alphanumeric	Status of escalation (None/Pending/Resolved)	Y

Sample Report Requirements

Usage Report	Escalation Report
Role Assistant Name	Conversation ID
Conversation Count	Role Assistant Name
Top Intents Handled	Escalation Reason
Average Confidence Score	Assigned Human Agent
Escalation Rate	Escalation Timestamp
Date Range	Resolution Timestamp

Appendix C: Requirement Traceability Matrix

This matrix will be populated progressively during Phases 2 and 3 of the project, linking each requirement to its architecture reference, design reference, code file, and test case identifiers.

Sl.No	Req. ID	Brief Description	Arch. Ref.	Design Ref.	Code Ref.	Test Case ID	
1	REQ-1.1	User registration	<TBD>	<TBD>	<TBD>	<TBD>	<
2	REQ-3.1	Voice query capture	<TBD>	<TBD>	<TBD>	<TBD>	<
3	REQ-4.1	Intent recognition	<TBD>	<TBD>	<TBD>	<TBD>	<
4	REQ-5.2	Workflow execution	<TBD>	<TBD>	<TBD>	<TBD>	<
5	REQ-9.1	Escalation to human agent	<TBD>	<TBD>	<TBD>	<TBD>	<